Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$
\begin{aligned}
a^{\mathbf{4}}_{ij} &= \tfrac{1}{g}\sum_{\gamma=1}^{r} r_\gamma \chi_\gamma^{R_M\oplus\mathbf{1}^2}\chi_\gamma^{(i)}\chi_\gamma^{(j)*}\\
a^{\mathbf{6}}_{ij} &= \tfrac{1}{g}\sum_{\gamma=1}^{r} r_\gamma \chi_\gamma^{R_M\oplus R_M\oplus\mathbf{1}^2}\chi_\gamma^{(i)}\chi_\gamma^{(j)*}
\end{aligned}
$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$
\left[\begin{array}{cccc}
g_2+{w_2}^2h_4+{n_2}^2h_5 & w_2w_3h_4+n_2n_3h_5 & w_2h_4 & n_2h_5\\
w_2w_3h_4+n_2n_3h_5 & g_3+{w_3}^2h_4+{n_3}^2h_5 & w_3h_4 & n_3h_5\\
w_2h_4 & w_3h_4 & h_4 & 0\\
n_2h_5 & n_3h_5 & 0 & h_5
\end{array}\right],
$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$
\begin{aligned}
&D^{++}=u^{+\alpha}\tfrac{\partial}{\partial u^{-\alpha}}; \qquad D^{--}=u^{-\alpha}\tfrac{\partial}{\partial u^{+\alpha}}\\
&2D^{++}=[D^0,D^{++}]; \qquad -2D^{--}=[D^0,D^{--}]\\
&D^0=[D^{++},D^{--}]=u^{+\alpha}\tfrac{\partial}{\partial u^{+\alpha}}-u^{-\alpha}\tfrac{\partial}{\partial u^{-\alpha}}
\end{aligned}
$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$
\begin{aligned}
A_N &= A_1(w_{N-1})\,A_{N-1} = \prod_{i=0}^{N-1} A_1(w_{N-i-1})\\
D_N &= D_1(w_{N-1})\,D_{N-1} = \prod_{i=0}^{N-1} D_1(w_{N-i-1})\\
beginalign*0.3em]C_N &= C_1(w_{N-1})\,A_{N-1} + D_1(w_{N-1})\,C_{N-1}\\
&= \sum_{i=0}^{N-1} D_i(w_{N-1-i})\,C_1(w_{N-1-i})\,A_{N-1-i}.
\end{aligned}
$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$u_i^{35,35}(y,z) = \left\{ \begin{array}{rcr} -5b_i^{35,35}(y,z) & if & |5b_i^{35,35}(y,z)| \leq 1, \\ -1 & if & -5b_i^{35,35}(y,z) < -1, \\ 1 & if & 5b_i^{35,35}(y,z) > 1, \end{array} \right\}, \quad i = 1,2,$$